Comprehensive Report: Driving Simulation Telemetry Dashboard

The Driving Simulation Telemetry Dashboard Application stands as a cutting-edge research and performance tool designed specifically for the world of high-fidelity virtual racing and vehicle dynamics analysis. Its primary purpose is to bridge the gap between the raw, often inaccessible physics data generated by modern simulators like BeamNG.drive and Assetto Corsa and the end-user—whether that user is a competitive sim-racer, a vehicle dynamics engineer, or a researcher studying human-machine interaction. By providing a sophisticated interface for real-time monitoring and post-session analysis, the application transforms a simple gaming experience into a data-driven laboratory. It serves not only as a digital dashboard for active driving but as a comprehensive flight recorder that captures every nuance of the vehicle's movement and the driver's intent.

The core functionality of the application is centered around its feature-rich live dashboard, which provides an immediate and visceral connection to the vehicle's state. Beyond standard instrumentation like the high-precision tachometer and speedometer, the dashboard incorporates advanced analytical modules such as the Friction Circle and the Input Visualizer. The Friction Circle is particularly critical, as it maps the interplay between lateral and longitudinal G-forces, allowing a driver to visually assess whether they are utilizing the "traction circle" of their tires to its fullest extent during transitions between braking and cornering. Simultaneously, the Input Visualizer provides millisecond-accurate feedback on throttle, brake, steering, and clutch applications, which is essential for identifying habits like "pedal overlap" or inconsistent steering inputs that can lead to instability. The application further extends its reach into vehicle longevity through a real-time health monitor that tracks wear and damage across five major systems—engine, transmission, suspension, brakes, and aerodynamics—alongside individual tire temperature and wear states. This granular level of detail ensures that every mechanical stress is accounted for during a run.

One of the most innovative aspects of the dashboard is its internal Simulation Mode, which operates independently of any external game engine. This mode is designed for testing, development, and research, allowing users to select from a variety of distinct "Driving Behaviors" that simulate different human profiles. These profiles range from the surgical precision of a "Professional" driver to the erratic and unpredictable patterns of "Drunk," "High," or "Reckless" operators. By introducing procedural noise, delayed reaction times, and swerving into the steering and pedal logic, the simulation mode creates a realistic environment for studying the impact of impairment or aggression on vehicle safety and control. This is complemented by a standalone Reaction Test module, which measures the driver's response time to visual stimuli using the physical brake pedal, providing a quantitative metric for fatigue or skill assessment.

When a session concludes, the application’s power shifts toward post-drive analysis and data portability. The internal database stores every frame of telemetry, which can then be reviewed through interactive time-series graphs or exported into a standardized CSV format for external processing. The exported data contains an exhaustive dictionary of variables that provide a complete reconstruction of the drive. Core variables such as `speed`, `rpm`, and `gear` provide the baseline for performance, while the high-fidelity `gForceX` (lateral), `gForceY` (longitudinal), and `gForceZ` (vertical) measurements allow for deep analysis of cornering forces and suspension travel. Advanced derived variables like `slip_angle_estimate`, `oversteer_correction`, and `understeer_plough` offer immediate insights into vehicle balance, helping engineers understand whether a car is sliding or pushing through a turn. These metrics are crucial for fine-tuning vehicle setups or correcting driver errors that are invisible to the naked eye.

The use cases for the exported CSV data are vast, particularly when examining the behavioral and health-related flags included in the log. Variables like `coasting_time_pct` and `is_coasting` are invaluable for fuel-saving analysis in endurance racing, while `is_wots` (Wide Open Throttle) and `pedal_overlap` help quant ify driver aggression and technical flaws in footwork. For safety research, the `jerk_x` and `jerk_y` variables—the rate of change of G-forces—serve as a proxy for driving "smoothness," where high jerk values correlate with erratic or unstable control. Furthermore, the tracking of `damage_` variables across different behavioral profiles allows for a direct correlation between driving style and mechanical failure rates. Ultimately, this application provides a holistic view of the driving experience, turning raw telemetry into actionable intelligence that can be used to improve performance, enhance safety, and deepen our understanding of the complex relationship between the driver and the machine.